3-TIER CLOUD ARCHITECTURE DEPLOYMENT ON AWS

A Production-Ready, Secure, and Highly Available Web Application

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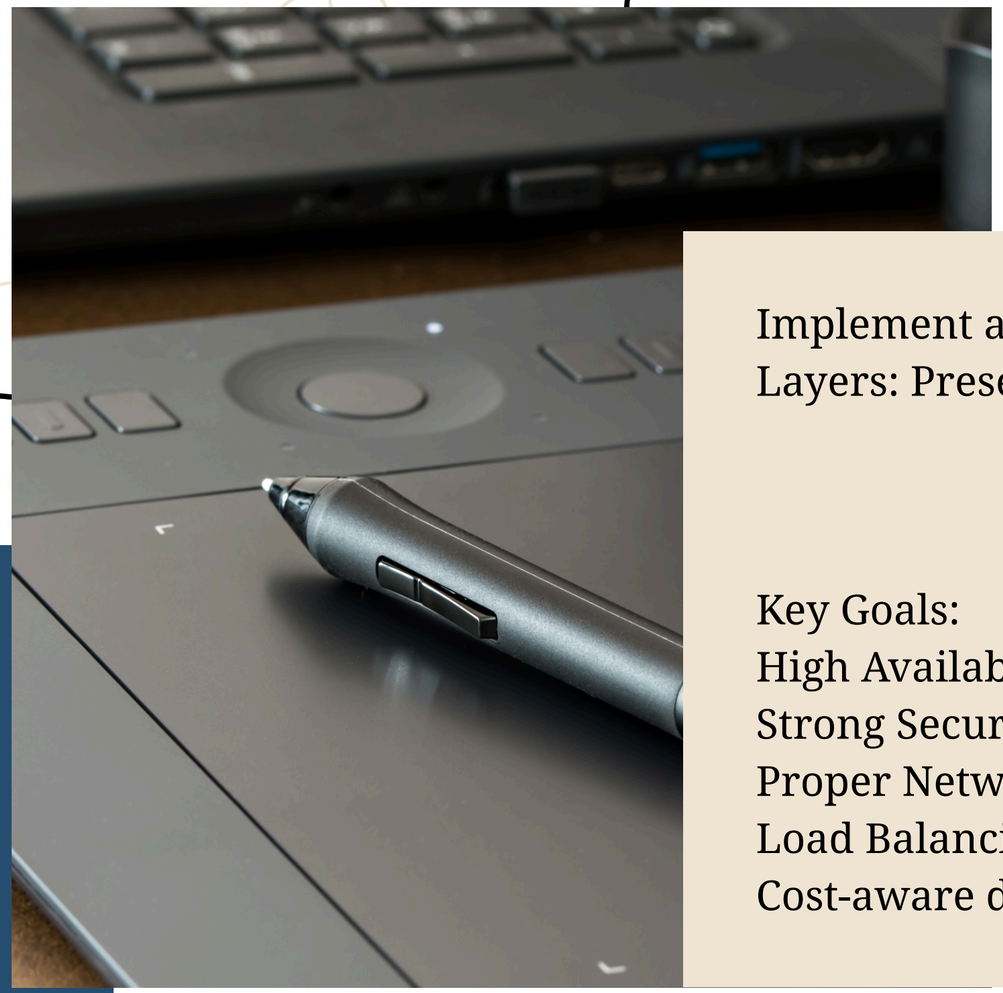
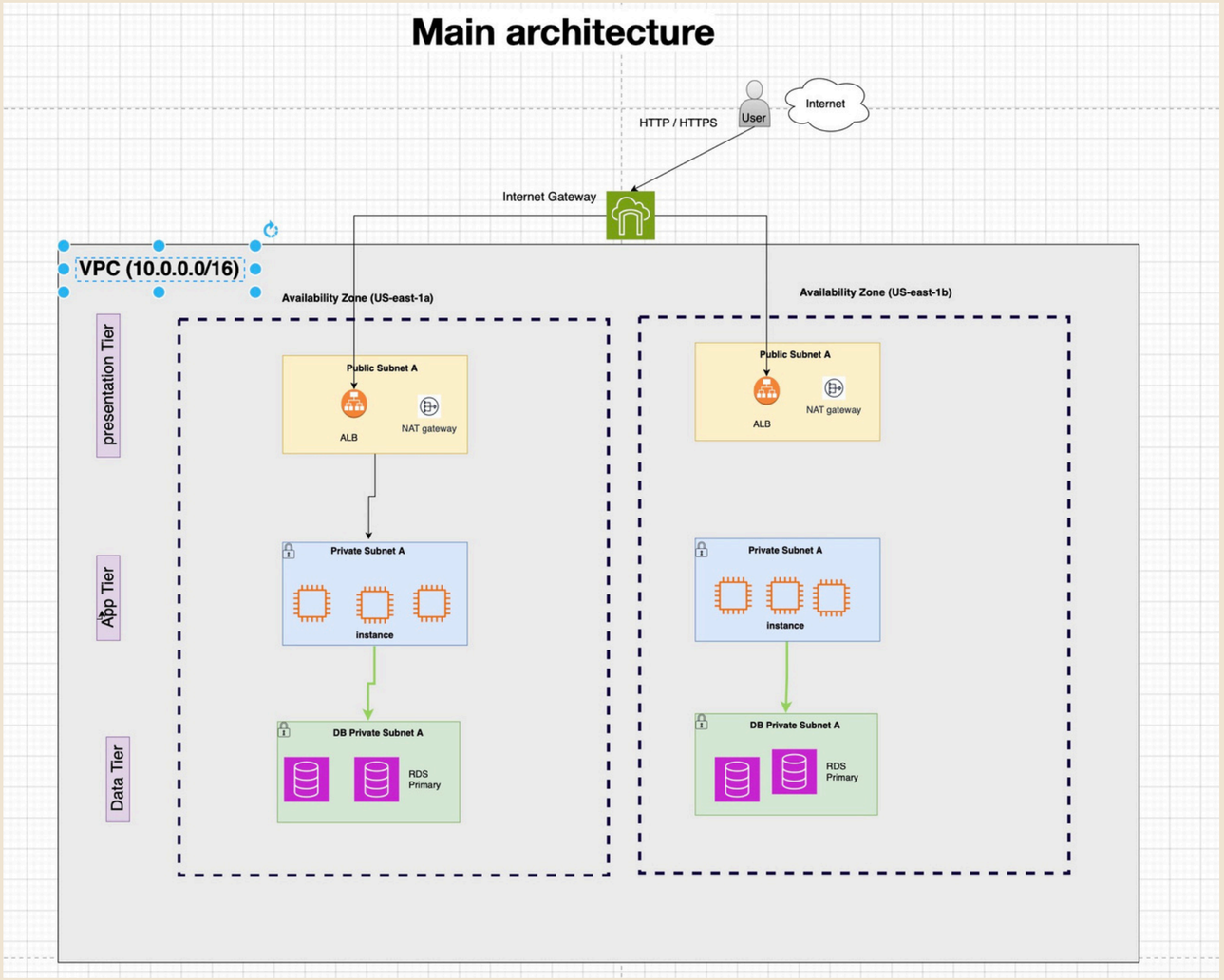
Project Overview



Implement a complete 3-tier web application architecture on AWS
Layers: Presentation (Public), Application (Private), Data (Private)

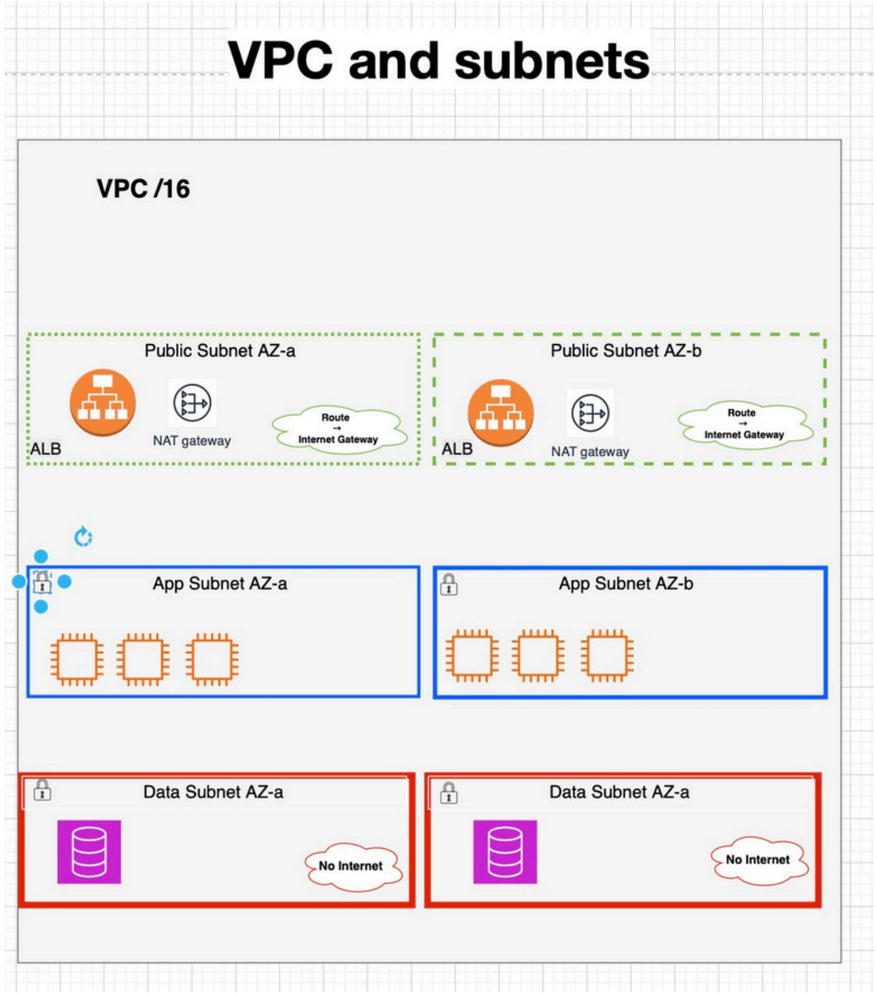
Key Goals:
High Availability across 2 Availability Zones
Strong Security with least privilege principle
Proper Network Segmentation (VPC + Subnets)
Load Balancing & Health Checks
Cost-aware design

High-Level Architecture



- Presentation Tier (Public Subnets) → ALB + NAT Gateway
- Application Tier (Private Subnets) → EC2 Instances
- Data Tier (Private Subnets) → RDS Multi-AZ

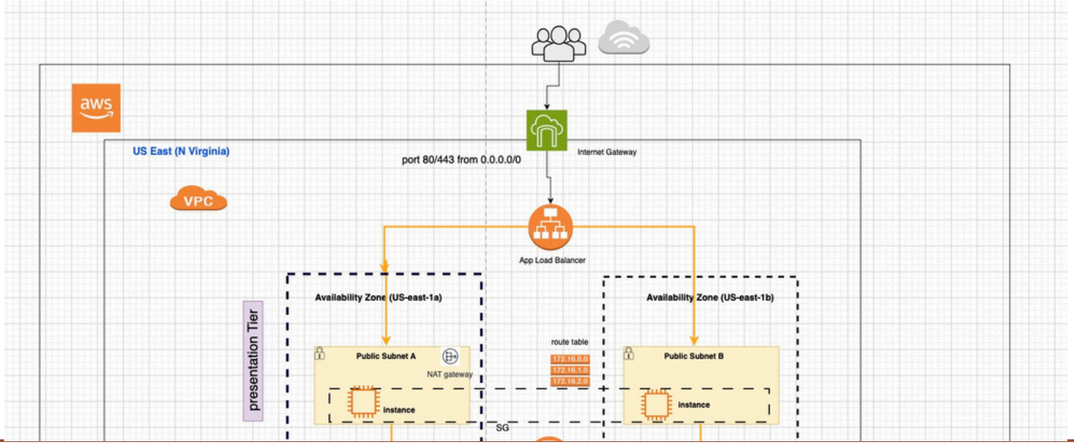
Network Infrastructure – VPC & Subnets



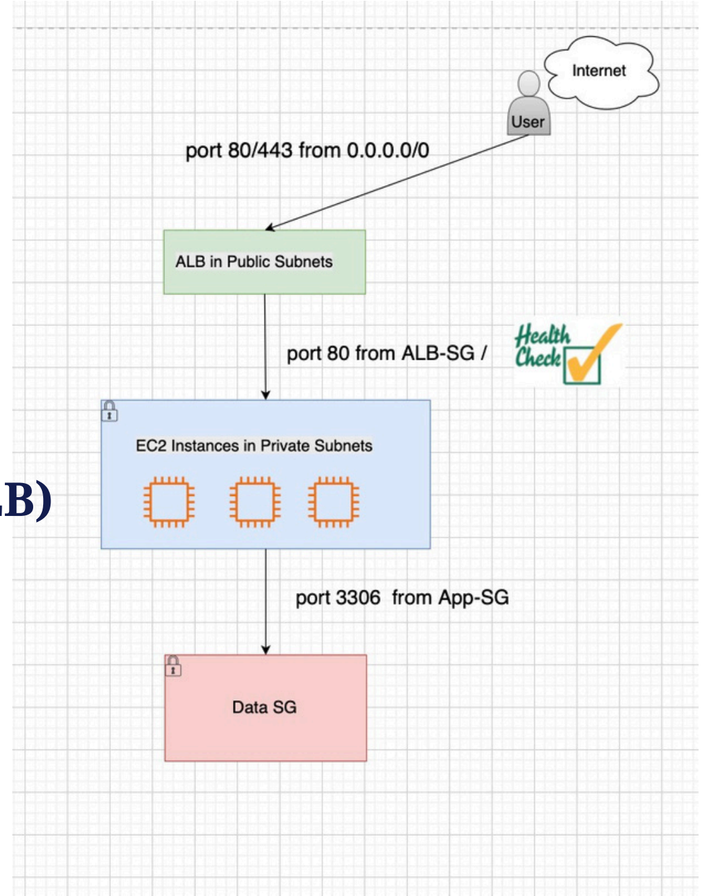
- VPC CIDR block: /16 for scalability
- 6 subnets across 2 Availability Zones
- Public subnets for ALB
- Private subnets for application tier
- Isolated subnets for data tier
- NAT Gateway enables outbound internet for private instances

Internet Access – NAT & IGW

- Internet Gateway attached to VPC → allows public subnets to access internet
- NAT Gateway placed in Public Subnets (us-east-1a & 1b)
- Private subnets route 0.0.0.0/0 → NAT Gateway
- Purpose: Allow outbound internet access from EC2 & RDS (updates, packages) without exposing them



Load Balancer & Application Tier



- **Internet-facing Application Load Balancer (ALB)**

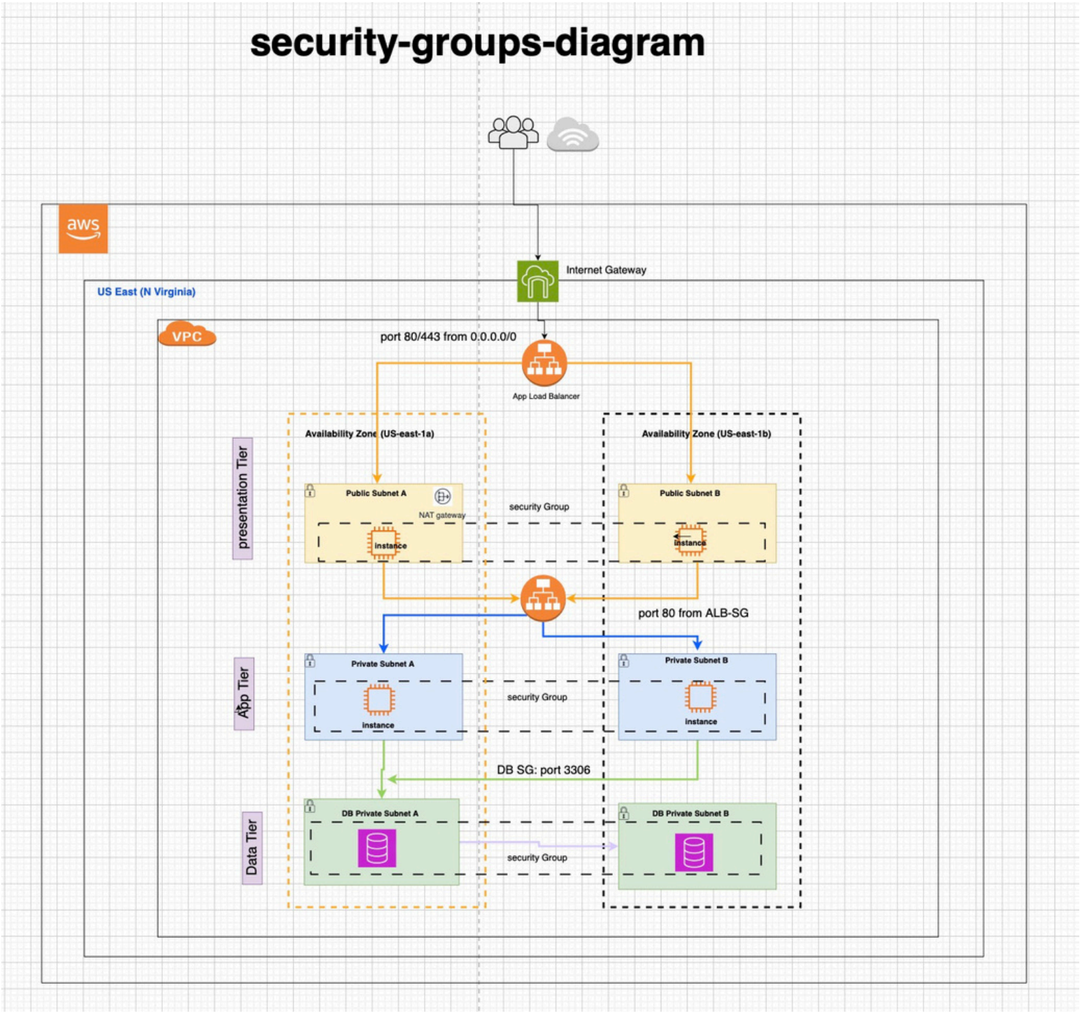
1. **Deployed in 2 public subnets**
2. **Listener: HTTP port 80**
3. **Health check path: /health**

- **3 EC2 instances in private app subnets**

1. **Registered in ALB target group**
2. **Display: Instance ID, AZ, DB connection status**

- **Traffic flow: User → ALB → EC2 instances**

Data Tier & Security Groups

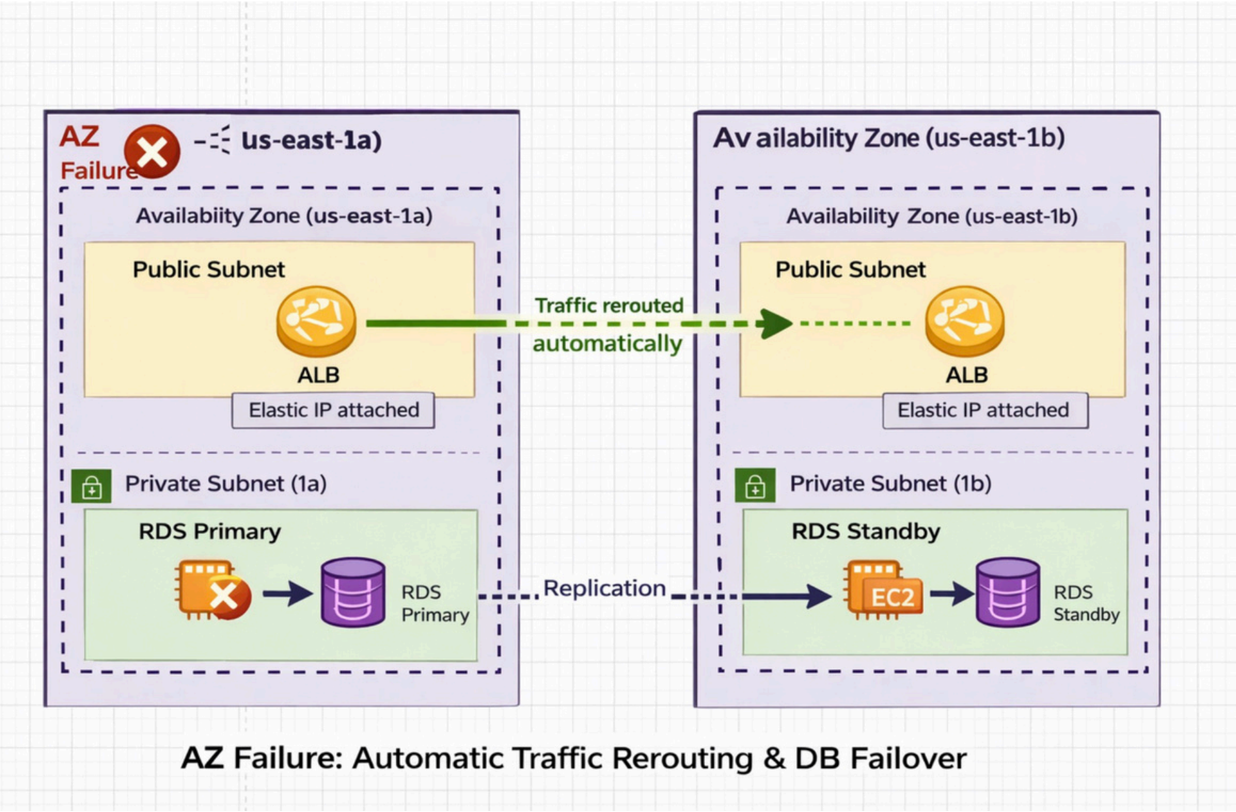


- RDS MySQL in private DB subnets
- Multi-AZ for high availability
- Security Groups – Least Privilege:

1. ALB-SG: Allow 80/443 from 0.0.0.0/0
2. App-SG: Allow 80 from ALB-SG only
3. DB-SG: Allow 3306 from App-SG only

- No direct internet access to private tiers

High Availability Features



- Multi-AZ deployment (2 Availability Zones)
- ALB distributes traffic across AZs
- Health checks ensure only healthy instances receive traffic
- RDS Multi-AZ with automatic failover
- If one AZ fails → traffic automatically routed to the other AZ

Design Decisions & Trade-offs

Decision	Professional	Cons
RDS (managed)	Easy Multi-AZ, backups	Slightly higher cost than EC2
Single NAT Gateway	Lower cost	Potential single point of failure
HTTP (no HTTPS)	Quick setup	Less secure (no encryption)
t3.micro instances	Free Tier eligible	Limited performance for heavy load

Cost Estimation

Resource	Approx. Hourly Cost	Notes
3 × t3.micro	~$0.03	Free Tier eligible
ALB	~$0.0225	Low usage
NAT Gateway	~$0.045	Most expensive item
RDS db.t3.micro	~$0.02	Multi-AZ adds ~50%
Total (24/7)	~$20–30/month	Shut down after testing!

Improvements & Next Steps

- Implement Auto Scaling Group
- Add RDS Multi-AZ database
- Add CloudWatch monitoring and alarms
- Consider multi-region disaster recovery

Conclusion & Q&A

- Built secure, highly available 3-tier architecture
- Learned: VPC segmentation, security groups, load balancing, cost awareness
- Ready for production-like environment with minor enhancements (HTTPS, Auto Scaling, CI/CD)

THANK YOU